

PERTH PEARCE, Australia

WMO: 946120

Lat: 31.6669S

Lon: 116.0189E

Elev: 54

StdP: 100.68

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.8	5.1	-2.5	3.1	23.5	-0.8	3.5	22.2	11.4	16.2	10.3	15.9	1.1	0	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.7	38.4	19.5	36.2	19.3	34.6	19.0	22.9	31.0	21.6	30.4	20.7	29.6	5.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.5	25.6	19.1	14.0	23.8	18.1	13.1	23.3	68.2	31.4	63.3	30.4	60.0	29.3	36.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.5	8.7	1.0	42.3	1.4	1.6	0.0	43.5	-0.9	44.4	-1.7	45.3	-2.7	46.5
			0.5	24.8	1.5	2.7	-0.6	26.8	-1.5	28.3	-2.3	29.8	-3.4	31.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	25.0	25.3	23.3	19.9	16.5	14.1	13.0	13.3	14.4	16.9	20.4	22.9
	DBStd	5.42	3.62	3.51	3.74	3.39	2.90	2.34	2.31	2.22	2.42	3.10	3.52	3.83
	HDD10.0	7	0	0	0	0	0	1	4	1	1	0	0	0
	HDD18.3	757	1	1	3	22	70	127	167	158	122	66	17	4
	CDD10.0	3182	465	427	414	296	203	124	96	102	131	215	311	399
	CDD18.3	891	207	194	159	67	14	1	0	1	3	23	78	144
	CDH23.3	9866	2404	2213	1667	584	115	2	0	4	37	259	886	1693
	CDH26.7	4807	1300	1170	793	197	19	0	0	0	6	79	387	857
Wind	WSAvg	4.2	5.0	4.9	4.6	3.7	3.5	3.6	3.6	3.6	3.8	4.2	4.7	4.9
Precipitation	PrecAvg	660	5	13	15	34	89	136	134	104	69	37	24	7
	PrecMax	970	37	97	86	114	196	289	243	180	136	84	94	38
	PrecMin	459	0	0	0	0	4	21	54	30	8	2	0	0
	PrecStd	124	9	20	18	26	43	53	47	32	32	20	22	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	40.5	39.0	34.0	28.9	23.2	21.9	23.7	27.0	32.4	36.9	40.1
		MCWB	20.2	20.4	18.7	17.5	15.5	13.7	13.6	14.4	15.7	17.3	17.7	19.3
	2%	DB	38.1	37.6	36.0	31.1	26.2	21.9	20.0	21.2	23.9	28.8	33.8	37.0
		MCWB	19.8	20.1	19.5	17.2	15.1	14.0	13.6	14.2	15.0	16.6	17.6	18.8
	5%	DB	35.9	35.3	33.2	28.9	24.1	20.2	18.8	19.8	21.6	26.1	31.0	34.1
		MCWB	19.5	19.7	18.6	17.1	14.9	13.6	13.3	13.8	14.7	16.1	17.3	18.7
	10%	DB	33.4	33.2	31.1	26.8	22.2	19.0	17.5	18.2	19.9	23.8	28.6	31.4
		MCWB	19.2	19.4	18.5	16.6	14.8	13.2	12.9	13.2	14.0	15.7	16.7	18.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	24.2	23.0	20.7	19.2	17.4	16.8	17.1	17.6	19.4	20.6	28.5
		MCDB	30.2	32.7	31.4	25.9	22.5	19.8	18.8	20.1	23.1	27.3	30.0	28.5
	2%	WB	22.5	22.7	21.5	19.7	18.0	16.3	15.4	15.7	16.4	18.1	19.4	21.5
		MCDB	31.2	31.4	28.9	25.4	21.5	18.9	17.7	18.7	21.2	25.2	28.4	30.4
	5%	WB	21.3	21.4	20.5	18.8	17.1	15.3	14.5	14.9	15.6	17.1	18.5	20.0
		MCDB	30.5	30.9	27.8	24.6	21.1	18.0	17.0	18.2	19.8	23.6	27.1	30.3
	10%	WB	20.3	20.5	19.6	18.0	16.2	14.4	13.7	14.0	14.7	16.3	17.7	19.0
		MCDB	30.0	30.1	27.4	24.1	20.2	17.2	16.3	17.2	18.8	22.0	25.8	29.0
Mean Daily Temperature Range	5% DB	MDBR	15.1	14.7	13.8	12.8	11.4	9.8	9.4	10.1	10.9	12.8	14.2	14.8
		MCDBR	18.2	17.3	16.3	15.2	13.8	11.3	10.4	12.2	14.1	16.2	18.1	18.5
		MCWBR	5.8	5.4	5.3	5.8	6.5	6.4	6.2	7.1	7.5	7.2	6.5	5.9
	5% WB	MCDBR	15.1	14.6	13.2	11.9	10.4	9.0	8.3	9.7	11.8	13.8	15.7	16.1
		MCWBR	5.8	5.4	5.3	5.7	6.2	6.3	5.9	6.3	7.0	6.8	6.4	6.2
Clear-Sky Solar Irradiance	taub	0.340	0.342	0.325	0.316	0.306	0.297	0.296	0.304	0.325	0.327	0.328	0.329	
	taud	2.558	2.569	2.605	2.616	2.616	2.625	2.624	2.594	2.510	2.526	2.548	2.564	
	Ebn at Noon	999	978	962	919	879	864	882	919	945	982	1005	1013	
	Edh at Noon	108	103	94	83	74	69	72	83	101	107	108	108	
All-Sky Solar Radiation	RadAvg	8.21	7.41	6.14	4.52	3.39	2.79	2.95	3.81	5.06	6.56	7.79	8.41	
	RadStd	0.38	0.26	0.30	0.26	0.23	0.19	0.18	0.18	0.28	0.29	0.32	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.51	N/A	N/A	+1.17	N/A	N/A	N/A	N/A	+194	N/A
Regional (18 neighbors)	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page